

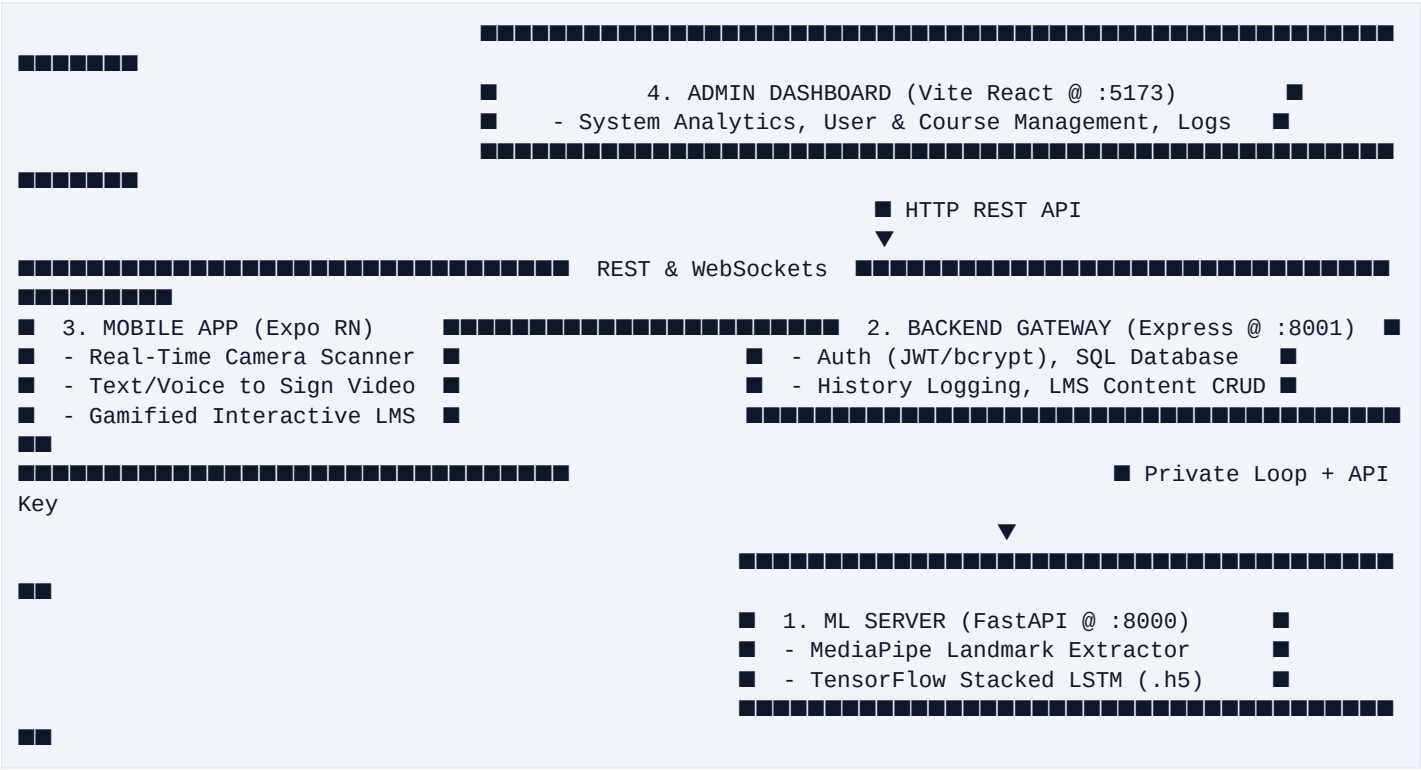
# ■ SignLens: Real-Time Sign Language Translation & Gamified Learning Ecosystem

## Master System Installation & Execution Guide for Academic Evaluators & Supervisors

Welcome to the **SignLens** research project repository! This document serves as a comprehensive, step-by-step guide designed specifically for academic supervisors, external evaluators, and non-developer users to run the full SignLens ecosystem locally from scratch.

### ■ 1. System Architecture Overview

SignLens is a decoupled, multi-tier microservices platform. The system consists of **four primary modules** that run together on your local machine:



### ■ 2. Prerequisite Software Installation Guide (For Beginners)

Before starting the application, ensure the following **four software packages** are installed on your host computer:

#### Step 2.1: Install Node.js (JavaScript Runtime)

- Download **Node.js v18 LTS** or **v20 LTS** from the official website: [https://nodejs.org/](https://nodejs.org/)
- Run the installer and click **Next** through all steps. Ensure **"Add to PATH"** is checked.
- Verify installation by opening **Command Prompt (cmd)** and running:

```
node -v
npm -v
```



```
# On Windows:
venv\Scripts\activate
# On Mac/Linux:
source venv/bin/activate

# 4. Install ML dependencies
pip install -r requirements.txt

# 5. Start Python FastAPI Server
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

■ **Success Check:** Open <http://localhost:8000/docs> in your browser. You should see the interactive FastAPI Swagger interface.

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### STEP 3: Launch Node.js API Gateway Backend (Port 8001)



Open a second Command Prompt window and execute:

```
# 1. Navigate to backend directory
cd backend

# 2. Install dependencies
npm install

# 3. Create .env file (if not present)
# Windows PowerShell command:
copy .env.example .env

# 4. Execute database migrations and seed default data
npm run migrate
npm run seed

# 5. Start Backend Gateway
npm run dev
```

■ **Success Check:** Open <http://localhost:8001/api-docs> in your browser. You should see the Node.js API Swagger documentation.

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### STEP 4: Launch Admin Web Dashboard (Port 5173)



Open a third Command Prompt window and execute:

```
# 1. Navigate to admin directory
cd admin

# 2. Install dependencies
npm install
```

```
# 3. Start Vite dev server
npm run dev
```

■ **Success Check:** Open `http://localhost:5173` in your browser to view the Admin Dashboard.



## STEP 5: Launch Mobile Client Application



Open a fourth Command Prompt window and execute:

```
# 1. Navigate to frontend directory
cd frontend

# 2. Install dependencies
npm install

# 3. Start Expo development server
npx expo start
```

■ **Testing Options:**

- **On Physical Phone:** Scan the displayed QR Code using your phone camera (iOS) or the **Expo Go** app (Android). (Ensure your phone and PC are connected to the same Wi-Fi network).
- **On Web Browser:** Press `w` in the terminal to launch the mobile layout in Google Chrome.
- **On Android Emulator:** Press `a` in the terminal (requires Android Studio installed).

## ■ 4. Component README Directory Map

For specialized, deep-dive instructions for each individual module, refer to the dedicated README files in their respective folders:

| Module Folder | Purpose                                                                                  | Link                                                                                             |
|---------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| model/        | Machine Learning model training scripts, datasets, MediaPipe extraction & FastAPI server | [model/README.md](file:///c:/Users/JOSHUA%20ASEMANI/Music/SignLens/SignLens/model/README.md)     |
| backend /     | Node.js Express Gateway proxy, MySQL/PostgreSQL ORM schemas, JWT auth & REST endpoints   | [backend/README.md](file:///c:/Users/JOSHUA%20ASEMANI/Music/SignLens/SignLens/backend/README.md) |

|           |                                                                                       |                                                                                                    |
|-----------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| frontend/ | React Native (Expo SDK 54) mobile camera scanner, voice/text to sign & gamified LMS   | [frontend/README.md](file:///c:/Users/JOSHUA%20ASEMANI/Music/SignLens/SignLens/frontend/README.md) |
| admin/    | Vite React administrative management web dashboard for analytics & LMS course editing | [admin/README.md](file:///c:/Users/JOSHUA%20ASEMANI/Music/SignLens/SignLens/admin/README.md)       |

## ■ 5. Comprehensive Troubleshooting & Common Issues Guide

### ■ Problem 1: "Python is not recognized as an internal or external command"

- **Cause:** Python was installed without enabling "Add to PATH".
- **Solution:** Uninstall Python, re-run the Python 3.10 installer, and check "**Add python.exe to PATH**" on the first screen. Restart Command Prompt.

### ■ Problem 2: ECONNREFUSED or Database Error when starting Backend

- **Cause:** MySQL / XAMPP or PostgreSQL database server is not running, or database `signlens_mobile` was not created.
- **Solution:**
  - Open XAMPP Control Panel and ensure **MySQL** is started.
  - Visit `http://localhost/phpmyadmin` and verify `signlens_mobile` database exists.
  - Re-run `npm run migrate` in `backend/`.

### ■ Problem 3: Port Conflict Error (EADDRINUSE: port 8000 or 8001 in use)

- **Cause:** Another program or zombie node/python process is occupying the port.
- **Solution (Windows Command Prompt):**

```
# Find process ID on port 8000 or 8001
netstat -ano | findstr :8000
# Kill process by ID (replace <PID> with number found)
taskkill /PID <PID> /F
```

### ■ Problem 4: Mobile App says "Network Error" when scanning signs

- **Cause:** The mobile app cannot reach your PC's local backend IP address.
- **Solution:**
  - Find your PC's local IP address by running `ipconfig` in Command Prompt (e.g., `192.168.1.100`).
  - Open `frontend/.env` and update:

```
EXPO_PUBLIC_API_URL=http://192.168.1.100:8001/api/v1
EXPO_PUBLIC_WS_URL=ws://192.168.1.100:8001/api/v1
```

- Restart Expo: `npx expo start -c`

## ■ 6. Project Authors & Supervision

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